Assignment 2

Marks 30

Due Date: September 21

1. As discussed in the class on Wednesday, August 19, determine experimentally or by simulation how the Fresnel's relations would work for unpolarized light incident on a glass surface (a microscope slide), i.e. how would reflectance and transmission change as a function of incident angle.
    a. If you would like to determine the answer experimentally, use light from a non-laser source (your smart phone torch would work well, but it is better to use a collimated LED torch or some such thing), determine the angle of incidence by some method, and measure the reflected light again using your phone itself as a detector. Download the software ImageJ, and find out the image intensity for every angle of incidence. Plot your results.
    b. If you would like to do a simulation, you can use Lumerical available in the Light Matter lab or COMSOL in the BioNap lab (let me know if you would like to use COMSOL), and set up an experiment. Analyse the results by plotting intensity versus angle.

    Explain your results quantitatively, and with physical arguments.